Summary of Self-Inductance

1- **Definition of Self-Induction: This is an effect where a changing magnetic flux through a circuit, and the resulting induced emf, arise from the circuit itself.**
2- **Definition of an RL Circuit:** A circuit containing a battery, a resistor, and an inductor.
3- **Definition of the Time Constant (τ):** The time interval required for the current in the circuit to reach 63.2% of its final value (ϵ/R). It is calculated as $\tau=L/R$.
4- Define the following terms:

a) **Kirchhoff's Loop Rule for RL Circuits**
b) **Current as a function of time**
c) The total energy U stored in an inductor
d) Magnetic Energy Density
e) The angular frequency of oscillation

5- Why does the current in a circuit not immediately jump to its maximum value ϵ/R when a switch is closed?

Answer: As the current increases, the magnetic flux through the circuit loop also increases, creating a back emf that opposes the battery's emf, resulting in a gradual increase to equilibrium.

6- Describe oscillations in an LC Circuit

Answer : It is a circuit containing a capacitor and an inductor where energy oscillates between the electric field of the capacitor and the magnetic field of the inductor.

7- An air-core solenoid containing 100 turns if the length of the solenoid is 20 cm and current passes through it 5 A. Determine the magnetic flux.

8- Calculate the energy stored in the magnetic field of a 50 m H inductor when it carries a steady current of 2 A?

9- An LC circuit consists of 20 m H inductor and 5pF capacitor. The capacitor is initially charged to 12 V before connected to inductor . a) find the angular frequency of oscillation .b) Calculate the maximum charge on the capacitor.

Lecture 6: Electricity and Magnetism II

1. Fundamental AC Concepts

- **AC Source:** Provides a time-varying voltage defined by $\Delta v = \Delta V_{max} \sin\omega t$, where ΔV_{max} is the voltage amplitude.
- **Frequency and Period:** The angular frequency is $\omega = 2\pi f$, where f is the frequency and $T = 1/f$ is the period.
- **Root Mean Square (rms):** Values used for practical power calculations, defined as $I_{rms} = I_{max}/\sqrt{2}$ and $\Delta V_{rms} = \Delta V_{max}/\sqrt{2}$.

2. Basic Circuit Elements

The behavior of current (i) and voltage (Δv) differs depending on the component:

Component	Instantaneous Voltage/Current Relation	Reactance/Resistance
Resistor	$\Delta v_R = I_{max} R \sin\omega t$	R (Resistance)
Inductor	$I_L = -\omega_L \Delta V_{max} \cos\omega t$	$X_L = \omega L$ (Inductive Reactance)
Capacitor	$I_C = \omega_C \Delta V_{max} \sin(\omega t + 2\pi)$	$X_C = \frac{1}{\omega c}$ (Capacitive Reactance)

3. RLC Circuits

Series RLC Circuits

- **Impedance (Z):** The total opposition to current, calculated as

  $Z = R^2 + (X_L - X_C)^2$.

- **Maximum Current:** $I_{max} = \frac{\Delta V max.}{Z}$

- **Phase Angle (ϕ):** The angle between current and voltage, where

  $\text{Tan}\phi = \frac{X_L - X_C}{R}$

- **Resonance:** Occurs when $X_L=X_C$, leading to

  The resonant frequency $\omega_0=\frac{1}{\sqrt{LC}}$

Parallel RLC Circuits

- **Voltage:** The instantaneous voltage is the same across all three elements.

- **Total Current:** $I=\sqrt{I_R^2+(I_L+I_C)^2}$

- **Impedance:** $\frac{1}{Z}=\sqrt{\frac{1}{R^2}+(\frac{1}{X_C}-\frac{1}{X_L})^2}$

- **Resonance:** Also occurs when $X_L=X_C$, resulting in $\omega_0=\frac{1}{\sqrt{LC}}$.

4. Power in AC Circuits

- **Average Power (P_{av}):** Calculated as $P_{av}=I_{rms}\Delta V_{rms}\cos\phi$
- **Power Factor:** The term cosϕ is known as the power factor.
- **Resistive Dissipation:** Power is only actually dissipated by the resistor, expressed as $P_{av}=I_{rms}^2 R$.

Problem 1: An 8μF capacitor is connected to the terminals of a 60Hz AC source whose rms voltage is 150 V. Find the capacitive reactance and the rms current in the circuit.

Solution:

Problem2: A series RLC AC circuit has R=425Ω, L=1.25 H, C = 3.5μF, , and V_{max} =150V.

a) Determine the inductive reactance, the capacitive reactance, and the impedance of the circuit.
b) Find the maximum current in the circuit.
c) Find the phase angle between the current and voltage.
d) Find both the maximum voltage and the instantaneous voltage across each element.

Solution:

Problem 3: Consider a series RLC circuit for which R=150Ω, L=20mH, ΔV_{rms}=20V, and . Determine the value of the capacitance for which the current is a maximum.

Solution: